Graduate Independent Study CSE5999

Washington University in St. Louis
James McKelvey School of Engineering

Student Name: Brian Zhou

Student ID: 608779

Date: 09/11/2026

Semester: Fall 2026

Topic:

Learning Multi-Modal Travel Speed from Remote Sensing and GPS Traces

Description:

This independent study investigates whether remotely sensed terrain and vegetation structure can substitute for trail attributes that a map does not have. Tobler's 1993 hiking function remains the default in GIS travel-time and accessibility modeling, yet it was fit to figures from a cartography textbook rather than to observed movement at scale. This project tests it against tens of thousands of real recorded hikes.

Deliverables: (1) a trajectory dataset built from OpenStreetMap GPS traces over one high-relief forested region, cleaned, stop-filtered, and map-matched to the trail graph, with per-segment observed speed, direction of travel, and inferred travel mode; (2) a remote sensing covariate stack of terrain derivatives from 1 m LiDAR, canopy cover, and vegetation structure from the point cloud; (3) a four-arm comparison of Tobler, OSM trail tags, remote sensing, and both, each predicting speed and variance.

The student brings computer vision research experience with two published papers (AIPMV 2024; ICPPOE 2023). Graded elements follow the course milestones: a project plan, progress updates, paper reviews, and a final report. The work is scoped toward a submission to IGARSS 2027, mentored by PhD students Danni Beaulieu and Tianyi Gao.

The program outlined in this proposal is adequate for 3 units of credit (maximum 6) in:

CSE5999 [x]

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Independent Study Advisor

Deliver Form to Computer Science Department by email: benos@wustl.edu or in person: McKelvey 1010